

Combinatorial Symmetric CV

Validating models on time-series presents unique challenges.

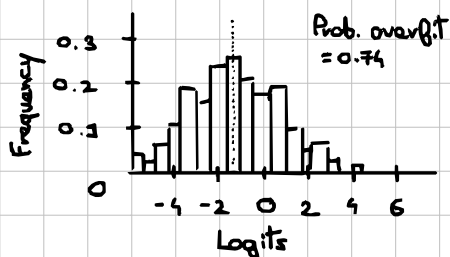
- Standard k-fold validation: fails when labels have serial conditionality as random sampling will place correlated or redundant events in training and testing, causing information leakage.
- Walk-forward method: while this method moves forward and prevents leakage, it lacks random sampling and relies on single testing path.

Estimating the Probability of Backtest Overfitting (PBO)

1. Form the performance matrix M of size $T \times N$.
Collects PnL across T observations and N model configs
2. Divide matrix M into an even number of disjoint matrices S , each of equal dimensions (M_S)
3. Generate all possible combinations (C_S) in groups of size $S/2$ (half of data) which is calculated with $\binom{S}{S/2}$.
4. Compare In-sample vs. Out-of-sample performances, for each C_S .
 - Join submatrices in training set J , and the rest in testing set $\bar{J}$.
 - Identify model with best perf. in training: $n^* = \text{argmax}(R_n)$
 - Determine the relative rank $\bar{w}_c$ of this model in testing: $\bar{w}_c = \frac{(\text{Rank } R_{n^*})}{N+1}$ ($\in [0, 1]$)
5. Calculate the logit $\lambda_c = \log\left(\frac{\bar{w}_c}{1-\bar{w}_c}\right)$. High logit indicate that OOS perf. is in line with IS perf. signaling low overfit.
6. Estimate PBO with $\int_{-\infty}^0 f(\lambda) d\lambda$, taking proportion of logit that signal underperformance ($\lambda < 0$) of OOS perf.

		Model variants				All possible combinations $\binom{S}{S/2}$			
		1	2	...	N				
Observations	1	0.1	-0.2	...		M_1	1	...	N
	2	-0.2	0.3	...		$\vdots$	0.3	...	
	$\vdots$					T/S	1		
	T	1.3	0.6	...		M_2	1	...	N
						$\vdots$	-0.3	...	...

Form J and $\bar{J}$ ($T/S \times S/2; N$)



Distribution of logits is skewed to negative values so $PBO > 0.5$.